

SIGN SIGHT: LEARNING EVALUATION AND BEHAVIOUR-AWARE ANALYSIS SYSTEM FOR TAMIL SIGN LANGUAGE

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

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ABSTRACT

The growing need for accessible communication technologies has increased the importance of sign language learning systems, particularly for deaf and mute individuals. While many existing solutions focus on sign recognition and translation, there is limited support for evaluating how effectively users learn and improve over time. This research presents the design and implementation of a Learning Evaluation and Behavior-Aware Analysis System as a dedicated component of a Tamil Sign Language learning platform.

The proposed component introduces a structured evaluation mechanism consisting of three learning levels: Basic, Intermediate, and Advanced. The Basic and Intermediate levels assess user knowledge through quiz-based interactions, while the Advanced level enables users to submit video-based answers. These video responses are evaluated by integrating outputs from a sign detection module, which identifies the performed sign and provides a confidence score. In addition to correctness evaluation, the system incorporates a behaviour monitoring mechanism to analyze user attention during video responses. A deep learning model based on MobileNetV2 is utilized to classify head direction from extracted video frames. The predictions are aggregated over time to identify attention patterns and detect possible distractions or suspicious behavior.

To improve the reliability of evaluation, a multi-factor scoring model is introduced by combining accuracy, behavior analysis, and confidence values. Furthermore, a prediction module is implemented to analyze user performance trends and classify learners into different levels. Based on these results, the system generates personalized feedback and recommendations to support continuous learning.

The proposed approach extends existing sign language learning systems by integrating evaluation, behavior analysis, and predictive capabilities into a single framework. The results demonstrate that the system provides a more comprehensive and effective method for assessing user learning in Tamil Sign Language, while enhancing engagement and overall learning experience.

Keywords: Tamil Sign Language, Learning Evaluation, Behavior Analysis, MobileNetV2, Computer Vision, User Performance Prediction, Sign Language Learning.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
TSL	Tamil Sign Language
CV	Computer Vision
API	Application Programming Interface
UI	User Interface
DB	Database
TF	TensorFlow

1. INTRODUCTION

In recent years, the rapid advancement of digital technologies has significantly influenced the development of modern learning systems. In particular, the integration of artificial intelligence, machine learning, and computer vision has enabled the creation of intelligent platforms capable of analyzing user interactions and providing personalized learning experiences. These technologies have transformed traditional learning methods into more interactive and data-driven systems, allowing learners to receive immediate feedback and improve their performance effectively.

Among the various domains that benefit from these advancements, sign language learning systems have become increasingly important. Sign language is a visual form of communication that relies on hand gestures, facial expressions, and body movements. It is widely used by individuals who are deaf or mute, making it an essential medium for communication. However, learning sign language is a complex process that requires continuous practice, proper guidance, and accurate evaluation. Unlike spoken language, sign language involves multiple visual components that must be interpreted correctly, which increases the difficulty of both learning and assessment.

Traditional methods of learning sign language, such as instructor-led training and recorded video tutorials, provide basic guidance but often lack interactive evaluation mechanisms. Learners using these methods may not receive immediate feedback on their performance, making it difficult to identify mistakes or measure improvement. As a result, the learning process becomes less efficient and may lead to slower skill development. This limitation has led to the development of technology-based solutions that aim to enhance sign language learning through automation and intelligent analysis.

Recent advancements in machine learning and deep learning have enabled the development of systems capable of recognizing sign language gestures with high accuracy. These systems typically utilize computer vision techniques to analyze hand movements and classify gestures into predefined categories. While such systems are effective in recognizing signs and supporting communication, they are primarily focused on detection and translation. As a result, they do not provide sufficient support for evaluating user learning or tracking progress over time.

In addition to recognition, effective learning systems must incorporate evaluation mechanisms that can assess user performance in a structured manner. Evaluation is a critical component of any learning process, as it helps learners understand their strengths and weaknesses. Without proper evaluation, it is difficult to ensure that users are improving their skills. However, most existing sign language systems do not include comprehensive evaluation frameworks, particularly for advanced learning scenarios involving video-based responses.

Another important aspect that is often overlooked in current systems is user behavior during learning activities. In video-based environments, factors such as attention, focus, and engagement play a significant role in determining learning outcomes. For example, a user who is distracted or not fully engaged may perform poorly, even if they understand the concepts. Therefore, monitoring user behavior can provide valuable insights into the learning process and improve the accuracy of evaluation. Despite its importance, behavior analysis is rarely integrated into sign language learning systems.

Furthermore, modern educational technologies emphasize the use of data-driven approaches to enhance learning experiences. Learning analytics and predictive models are widely used in various domains to analyze user performance and provide personalized recommendations. These approaches enable systems to identify learning patterns, predict user progress, and offer guidance for improvement. However, such techniques are not commonly applied in sign language learning platforms, especially in systems designed for Tamil Sign Language.

Tamil Sign Language, in particular, has limited technological support compared to other widely used sign languages. This creates a gap in accessible learning tools for Tamil-speaking users who rely on sign language for communication. Addressing this gap requires the development of systems that are specifically designed to support learning, evaluation, and improvement within this context.

Considering these limitations, there is a clear need for a comprehensive system that goes beyond simple gesture recognition and incorporates evaluation, behavior analysis, and predictive capabilities. Such a system can provide a more effective and reliable learning experience by assessing user performance, monitoring engagement, and offering personalized feedback.

Therefore, this research focuses on the design and development of a Learning Evaluation and Behavior-Aware Analysis System for Tamil Sign Language. The proposed component introduces a structured approach to evaluating user learning through multiple levels, including quiz-based assessment and video-based response validation. It also integrates a behavior analysis mechanism using deep learning techniques to monitor user attention during learning activities. In addition, a prediction module is implemented to analyze user performance trends and provide feedback for improvement.

By combining these features into a unified framework, the proposed system aims to enhance the effectiveness of sign language learning, improve evaluation accuracy, and support continuous user development.

1.1 Background and Literature Survey

The development of intelligent sign language learning systems has become an important research area due to the increasing need for accessible digital learning tools for deaf and mute communities. Sign language is not only a communication method but also a language system that depends on hand gestures, body movement, facial expression, direction, timing, and visual understanding. Therefore, building a technological solution for sign language is more complex than developing a normal text-based learning system. A proper sign language platform should not only recognize gestures but also support the learner throughout the learning process by giving feedback, measuring performance, identifying weak areas, and guiding improvement.

Most of the early and recent research in this area has mainly focused on sign language recognition. Recognition systems attempt to identify signs performed by users and convert them into text, speech, or another meaningful output. Perera et al. developed a Tamil alphabet sign language recognition system using MediaPipe and machine learning techniques [5]. Their work is highly relevant to the Tamil Sign Language domain because it focuses on Tamil signs rather than only widely researched sign languages. The use of MediaPipe in their study shows the importance of extracting structured hand landmark features before applying machine learning algorithms. Methods such as Random Forest and Support Vector Machines are useful when the input features are already well-defined. This type of approach is suitable for recognizing static or semi-static signs where hand positions can be represented through landmark coordinates. However, the main limitation is that the system is focused on recognition. It does not attempt to evaluate whether a learner is improving, whether the learner repeatedly makes mistakes, or whether feedback should be adapted based on performance. Therefore, while this type of work supports the recognition side of sign language technology, it does not fully address the learning evaluation problem.

Deep learning approaches have also been widely used to improve the performance of sign language recognition systems. Pandiyarajan et al. proposed a real-time sign language recognition and translation system using CNN and BiLSTM models [6]. This is important because sign language is often dynamic, and many signs cannot be understood from a single image frame. CNN models are useful for extracting spatial features from image frames, while BiLSTM models can learn temporal patterns from a sequence of frames. This makes the system more suitable for recognizing video-based sign language gestures. The strength of this approach is that it considers movement over time, which is important for real-world sign

language recognition. However, similar to many recognition systems, the purpose of this work is mainly communication accessibility and translation. It helps identify what sign is performed, but it does not evaluate the user as a learner. It does not answer questions such as whether the user's performance improved compared to previous attempts, which sign categories are difficult for the user, or what type of learning support should be recommended next.

Interactive sign language learning systems move one step closer to the educational side of the problem. Ravikumar et al. introduced an interactive sign language learning system using deep learning [2]. This research is strongly related to the proposed component because it focuses on sign language learning rather than only recognition. The system includes features such as real-time feedback, image recognition, word recognition, and learning interaction. These features are important because learners benefit when the system responds immediately to their actions. In a practical learning environment, real-time feedback can help users correct their mistakes quickly and remain engaged with the platform. However, even though this type of system improves interaction, it still does not fully solve the structured learning evaluation problem. Real-time feedback tells the learner whether a particular action is correct at that moment, but it does not necessarily provide a deeper analysis of performance over time. A complete learning evaluation system should store attempts, calculate scores, identify weak areas, measure progress trends, and predict the user's learning level. Therefore, Ravikumar et al.'s work provides a strong foundation for interactive learning, but there is still room to extend such systems with performance prediction and behavior-aware evaluation.

The Master-ASL system proposed by Cheemakurthi and Chen is another important reference because it directly combines sign language learning and assessment [1]. This is closer to the direction of the proposed component because assessment is an essential part of any learning environment. A system that only teaches signs without assessing the learner cannot clearly measure learning outcomes. Master-ASL shows that sign language platforms can be designed not only for practice but also for evaluation. This supports the idea that sign language learning should include structured assessment mechanisms. However, assessment alone is not enough for a complete intelligent learning system. If the evaluation only checks correctness, it may ignore other important factors such as user engagement, attention, confidence, and learning progress. For example, two learners may receive the same score, but one learner may be improving steadily while the other may be repeating the same mistakes. A more intelligent evaluation system should be able to distinguish these situations by analyzing historical

performance and learning trends. This is one of the areas where the proposed component extends beyond basic assessment.

Apart from sign language-specific research, learning analytics provides an important theoretical foundation for this component. Siemens and Baker explain learning analytics as the measurement and analysis of learner data to understand and improve learning processes [10]. This concept is highly relevant because the proposed system depends on user performance data such as quiz scores, accuracy, category-wise performance, response history, confidence values, and behavioural indicators. In a normal learning platform, these values may only be displayed as simple results. However, in a learning analytics-based system, the same data can be used to identify patterns and provide meaningful feedback. For example, if a user repeatedly performs poorly in one sign category, the system can identify that category as a weak area. If the user's score increases over multiple attempts, the system can detect positive progress. This type of analysis supports personalized learning because the feedback is based on actual user performance rather than a general recommendation.

AI-driven self-directed learning research further supports this direction. Younas et al. reviewed AI-based approaches used in self-directed learning and highlighted how AI can support learner autonomy, engagement, and adaptive learning experiences [3]. This is important because sign language learners may use the platform independently without a teacher always being present. In such a case, the system must guide the learner by providing feedback, recommendations, and progress indicators. A self-directed learner needs to know what to practice next, which areas are weak, and whether their performance is improving. Therefore, the proposed component uses score-based prediction and feedback generation to support independent learning. Instead of simply showing a final mark, the system interprets the result and provides guidance for improvement.

Intelligent tutoring systems also provide useful background for the proposed work. Khalaily discusses AI-powered intelligent tutoring systems and their role in providing personalized support to learners [4]. Intelligent tutoring systems are designed to act as a virtual tutor by adapting content, feedback, and guidance according to the learner's needs. This idea is relevant to the proposed component because the Learning Evaluation System aims to provide more than a fixed quiz result. It attempts to classify the learner's level, identify weak areas, and recommend improvement directions. In this sense, the component follows the idea of intelligent tutoring by using learner data to provide individualized feedback. However,

applying this concept to Tamil Sign Language requires additional visual and behavioural analysis because the learning activity involves video responses and physical gestures, not only text-based answers.

A major limitation in many existing sign language learning systems is that they do not consider user behavior during learning. In video-based learning, the user's attention and engagement can influence the quality of the response. For example, during an advanced-level video answer, a user may look away frequently, lose focus, or behave in a way that affects the reliability of the evaluation. In traditional online learning, behavior monitoring has been used to estimate attention and engagement. In computer vision research, facial behavior analysis provides techniques that can be adapted for this purpose. Baltrusaitis et al. introduced OpenFace as an open-source facial behavior analysis toolkit [7]. OpenFace demonstrates that facial landmarks, eye gaze, and head movement can be analyzed using computer vision techniques. Although OpenFace itself is not a sign language learning system, the concepts behind facial behavior analysis are useful for monitoring attention during video-based learning activities.

Facial landmark detection is also a key background area for behavior-aware learning systems. Wu and Ji provided a detailed survey of facial landmark detection methods [8]. Facial landmark detection is useful for identifying important facial points such as eyes, nose, mouth, and face contour. These landmarks can be used to estimate head direction, facial orientation, and attention-related behavior. In the proposed component, behavior monitoring is not intended to replace sign recognition. Instead, it supports evaluation by adding an attention-related layer. This is important because Mark's sign detection component focuses on whether the user performed the correct sign, while the behavior monitoring layer focuses on whether the user was attentive during the video response. This separation makes the proposed component more meaningful because it avoids duplicating the sign detection task and instead adds a different type of learning-related analysis.

The choice of deep learning architecture is also important for behavior analysis. MobileNetV2, introduced by Sandler et al., is a lightweight convolutional neural network architecture designed to reduce computational cost while maintaining good image classification performance [9]. This makes it suitable for real-time or near-real-time visual classification tasks. In the proposed system, MobileNetV2 can be used to classify head direction from video frames, such as straight, left, right, up, or down. These frame-level predictions can then be aggregated over time to produce an attention score. The reason for using a model such as

MobileNetV2 is that head direction classification involves visual patterns that may vary across users, lighting conditions, and camera angles. A purely fixed rule-based method may not generalize well in such situations. However, the final suspicious or attention-related decision should not depend on a single frame. Instead, the system should analyze patterns across the entire video response.

When all these areas are considered together, it becomes clear that the proposed component is built from multiple research foundations. Sign language recognition studies provide the base for understanding how user signs can be detected [5], [6]. Interactive learning systems show the importance of feedback during sign language learning [2]. Sign language assessment systems show the need for structured evaluation [1]. Learning analytics and intelligent tutoring research support the use of score-based prediction and personalized feedback [3], [4], [10]. Facial behaviour analysis and landmark detection provide the foundation for monitoring attention during video responses [7], [8]. MobileNetV2 supports the technical implementation of lightweight visual classification [9].

However, the key issue identified from the literature is that these areas are often handled separately. Recognition systems usually stop after detecting the sign. Learning systems may provide feedback but often do not deeply analyze long-term performance. Assessment systems may check correctness but may not consider attention or behavioral factors. Learning analytics systems are strong in prediction but are mostly applied to general education rather than sign language learning. Behavior analysis systems can monitor attention but are not commonly integrated into sign language learning platforms. This separation creates a clear research gap.

The proposed Learning Evaluation and Behavior-Aware Analysis System addresses this gap by combining these ideas into one component. The system evaluates Basic and Intermediate levels using quiz-based assessment, while the Advanced level uses video-based answer validation. For advanced video answers, the system receives the predicted sign and confidence score from the sign detection component and compares it with the expected answer. In parallel, the behavior monitoring model analyzes the user's head direction across video frames to estimate attention. The final evaluation is not based only on correctness but also considers confidence, attention, and previous performance trends. This allows the system to provide a more complete understanding of the learner's progress.

Therefore, the literature shows that the proposed component is not simply a quiz module or a gesture recognition module. It is an evaluation-focused learning component that connects recognition, behavior analysis, learning analytics, and prediction. This combination makes the system more suitable for a real learning environment, where the main goal is not only to recognize signs but also to help users improve their Tamil Sign Language skills over time.

1.2 Research Gap

The rapid advancement of artificial intelligence and computer vision technologies has significantly improved the development of sign language recognition and learning systems. However, a critical analysis of existing literature reveals that current solutions remain limited in addressing the full requirements of an effective and structured sign language learning environment. While several studies have contributed to gesture recognition, translation, and interactive learning, there are still notable gaps in integrating these capabilities into a comprehensive learning evaluation framework.

One of the primary gaps identified is the **lack of structured learning evaluation mechanisms** in existing systems. Many studies, such as the Tamil sign language recognition system proposed by Perera et al. [5], focus on accurately identifying gestures using machine learning techniques. Similarly, deep learning-based approaches such as CNN and BiLSTM models have demonstrated high accuracy in recognizing dynamic gestures [6]. However, these systems are primarily designed for recognition and translation purposes rather than for evaluating the learner's performance. They do not track user progress, identify learning difficulties, or provide structured feedback based on performance. As a result, these systems are limited in their ability to function as effective educational tools.

Another important gap is the **absence of performance prediction and learning analytics integration**. Learning analytics research highlights the importance of using learner data to analyze performance patterns and provide personalized feedback [10]. Furthermore, AI-driven learning systems emphasize adaptive learning and self-directed learning capabilities [3], [4]. Despite these advancements, most sign language learning platforms only provide immediate feedback, such as correct or incorrect responses, without analyzing historical data or predicting future learning outcomes. This lack of predictive capability limits the system's ability to support continuous learning and improvement.

A further limitation is the **lack of integration between sign recognition systems and learning evaluation frameworks**. Existing systems such as those proposed by Pandiyarajan et al. [6] and Ravikumar et al. [2] demonstrate strong capabilities in recognizing gestures and providing interactive feedback. However, these systems treat recognition as an independent process and do not utilize recognition outputs (such as predicted labels and confidence scores) for evaluating learner performance. This creates a disconnect between technical recognition

models and educational evaluation systems, reducing the overall effectiveness of the learning process.

In addition, most current systems **do not consider user behavior and attention during learning activities**. Behavioral analysis techniques, such as facial landmark detection and head movement tracking, have been widely explored in computer vision research [7], [8]. These techniques can provide valuable insights into user attention and engagement. However, such approaches are rarely integrated into sign language learning systems. As a result, evaluations are based solely on the final output (correct or incorrect gesture), without considering whether the learner was attentive or distracted during the task. This leads to incomplete and potentially inaccurate assessments of learner performance.

Another significant gap is the **lack of multi-level evaluation strategies in existing systems**. Most platforms evaluate user responses using a single approach, regardless of the complexity of the task. However, sign language learning involves different levels of difficulty, including basic recognition, intermediate understanding, and advanced application. Existing systems do not adapt their evaluation methods according to these levels, which reduces the accuracy and relevance of the assessment. For example, simple quiz-based answers and complex video-based responses are often evaluated using the same criteria, which is not suitable for a comprehensive learning environment.

Furthermore, there is a **limited use of lightweight and efficient deep learning models for real-time behavioral analysis** in sign language systems. While deep learning architectures have been widely used for gesture recognition, their application in behavior analysis within learning systems remains limited. MobileNetV2, proposed by Sandler et al. [9], provides an efficient solution for real-time image classification tasks with low computational cost. However, such lightweight models are not widely utilized in sign language learning platforms for behavior monitoring, particularly in web-based environments where performance and responsiveness are critical.

How the Proposed System Addresses These Gaps

The proposed Learning Evaluation System for Tamil Sign Language addresses these research gaps by introducing an integrated and intelligent evaluation framework. Unlike existing

systems, it combines recognition outputs, performance analytics, and behavioral analysis to provide a more comprehensive learning experience.

The system introduces a **multi-level evaluation framework**, where Basic and Intermediate levels are evaluated using structured quizzes, while the Advanced level uses video-based responses. This ensures that each level is assessed appropriately based on its complexity.

In addition, the system integrates **learning analytics and prediction mechanisms**, where user scores and historical performance data are analyzed to generate learning insights and predictions. This enables personalized feedback and supports self-directed learning.

The system also bridges the gap between recognition and evaluation by **utilizing outputs from the sign detection component**, including predicted labels and confidence scores, to evaluate user responses more effectively.

Furthermore, a **behavior-aware analysis module** is introduced, which uses a lightweight deep learning model (MobileNetV2) to analyze user attention during video-based tasks. This enhances the reliability of the evaluation by incorporating behavioral context.

Finally, the system is designed as a **real-time, web-based platform**, ensuring efficient performance and practical usability for learners.

Feature	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	Proposed System
Sign Language Recognition	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓	✓
Learning Evaluation	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓
Performance Tracking	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Behavior Analysis	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✓
Prediction / Analytics	✗	✗	✓	✓	✗	✗	✗	✗	✗	✓	✓
Multi-Level Evaluation	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Real-time Efficient Model	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓

Table 1: Research Gap Comparison

1.3 Research Problem

Although significant advancements have been made in sign language recognition and learning systems, existing solutions are still limited in supporting a complete and effective learning evaluation process. Many current systems are designed to recognize gestures and convert them into text or speech using machine learning and deep learning techniques [5], [6]. While these approaches improve communication accessibility, they do not adequately support the learning process, as they fail to evaluate how well a learner understands and improves over time.

One of the main issues is the absence of a **structured evaluation mechanism** within sign language learning platforms. In a typical learning environment, evaluation is essential for measuring user performance, identifying weak areas, and guiding improvement. However, most existing systems only provide basic outputs such as correct or incorrect responses without offering deeper insights into user performance. Although some platforms include limited assessment features [1], these are not sufficient to evaluate learning across different levels of complexity, such as basic recognition, intermediate understanding, and advanced application.

Another critical problem is the lack of **performance tracking and learning progression analysis**. Effective learning systems should be able to monitor user performance over time and identify trends that indicate improvement or decline. However, current sign language learning systems do not maintain detailed performance histories or analyze user progress. As a result, learners are unable to understand how their skills evolve, and the system cannot provide meaningful feedback based on past performance.

The **limited application of learning analytics and prediction techniques** is a significant issue in existing systems. Research in artificial intelligence highlights the importance of analyzing learner data to generate insights and predict future performance [3], [10]. Intelligent tutoring systems also emphasize adaptive learning, where the system adjusts its behaviour based on the learner's progress [4]. However, such capabilities are not commonly implemented in sign language learning platforms. Without prediction and analytics, the system cannot support personalized learning or provide recommendations tailored to individual users.

Another important problem is the **lack of integration between recognition models and evaluation frameworks**. Advanced models such as CNN and BiLSTM are capable of accurately detecting dynamic gestures [6]. However, the outputs of these models, including

predicted labels and confidence scores, are not effectively utilized for evaluating user responses. This creates a gap between technical recognition capabilities and practical learning applications, reducing the overall effectiveness of the system as a learning tool.

In addition, most existing systems do not consider **user behavior and attention during learning activities**, particularly in video-based scenarios. Behavior analysis techniques, such as facial landmark detection and attention monitoring, have been widely explored in computer vision research [7], [8]. These techniques can provide valuable information about user engagement, which is an important factor in the learning process. However, such approaches are rarely integrated into sign language learning systems. As a result, evaluations are based only on the correctness of responses, without considering whether the learner was attentive or engaged during the task. This can lead to incomplete and potentially inaccurate assessments.

Moreover, there is a challenge in implementing **efficient and real-time capable solutions** for behavior-aware learning systems. While deep learning models provide high accuracy, they often require high computational resources. Lightweight architectures such as MobileNetV2 have been proposed to address this issue by providing efficient performance suitable for real-time applications [9]. However, these models are not widely utilized in sign language learning platforms, particularly for behavior analysis in web-based environments.

Considering these limitations, it is evident that existing systems do not provide a comprehensive solution for evaluating sign language learning. They either focus on recognition, interaction, or analytics individually, without integrating these components into a unified framework. As a result, learners do not receive accurate, reliable, and personalized feedback that supports continuous improvement.

Therefore, the main problem addressed in this research can be defined as:

How to design and develop an integrated Learning Evaluation System for Tamil Sign Language that can accurately assess user performance, incorporate behavior analysis, and utilize learning analytics to provide predictive and personalized feedback within a structured and real-time learning environment?

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to design and develop an integrated Learning Evaluation System for Tamil Sign Language that can accurately assess user performance through structured multi-level evaluation, incorporate behavior-aware analysis during video-based responses, and utilize learning analytics to provide predictive and personalized feedback in a real-time learning environment.

1.4.2 Specific Objectives

- **To design a structured multi-level evaluation framework for Tamil Sign Language learning, including Basic, Intermediate, and Advanced levels**

This objective focuses on creating a well-organized evaluation structure that reflects different levels of learning complexity. It ensures that users are assessed appropriately based on their understanding, starting from basic recognition to advanced application using video-based responses.

- **To implement quiz-based evaluation mechanisms for assessing user performance in Basic and Intermediate levels**

This objective aims to develop interactive quiz modules that can measure user understanding through structured questions. These quizzes allow the system to evaluate correctness, accuracy, and consistency in a controlled manner.

- **To evaluate video-based user responses in the Advanced level by utilizing outputs from the sign detection component**

This objective ensures that advanced-level learning is assessed using real-world scenarios, where users submit video responses. The system uses outputs such as predicted labels and confidence scores from the sign detection module to validate user answers.

- **To develop a behavior-aware analysis module to monitor user attention during video-based responses using computer vision techniques**

This objective introduces an additional evaluation layer that considers user behavior during learning. By analyzing user attention, the system can improve the reliability of evaluation results.

- **To apply a lightweight deep learning model (MobileNetV2) for classifying user head direction and detecting attention patterns**

This objective focuses on implementing an efficient deep learning model capable of performing real-time classification of head direction (e.g., left, right, straight). This supports behavior analysis without affecting system performance.

- **To design a multi-factor scoring mechanism that combines accuracy, confidence, and behavior analysis for reliable evaluation**

This objective aims to enhance evaluation accuracy by combining multiple factors instead of relying only on correctness. It ensures that the final score reflects both performance and user engagement.

- **To implement performance tracking and learning analytics to analyze user progress over time**

This objective focuses on storing and analyzing user performance data across multiple attempts. It enables the system to identify patterns, measure improvement, and support data-driven evaluation.

- **To develop a prediction module that classifies user learning levels and provides personalized feedback and recommendations**

This objective aims to use performance data to predict the user's learning level. Based on this prediction, the system provides personalized feedback to guide users in improving their skills.

2. METHODOLOGY

This chapter describes the methodology followed to design and develop the Learning Evaluation and Behavior-Aware Analysis System for Tamil Sign Language. The proposed component is designed to evaluate user learning performance through structured assessment methods, analyze user behavior during video-based responses, and provide predictive feedback based on performance trends.

The methodology is developed with the objective of creating a system that not only validates user responses but also understands the learning process by considering multiple factors such as accuracy, confidence, and user attention. Unlike traditional systems that focus only on gesture recognition, this component integrates evaluation, behavior analysis, and learning analytics into a unified framework.

The overall methodology consists of several key stages, including system architecture design, data processing, evaluation mechanism development, behavior analysis, performance scoring, and prediction. Each stage is designed to address specific limitations identified in existing systems and to ensure that the final solution provides a comprehensive and reliable learning evaluation process.

The system operates through a multi-level evaluation structure. In the Basic and Intermediate levels, user knowledge is assessed using quiz-based interactions. These levels are designed to evaluate recognition and understanding through structured questions. In the Advanced level, users are required to submit video-based responses, which are analyzed using outputs from the sign detection component. This allows the system to evaluate real-world application of sign language skills.

In addition to correctness evaluation, the system incorporates a behavior-aware analysis mechanism. During video-based responses, the user's head direction is analyzed using a deep learning model to determine attention patterns. This provides additional context to the evaluation process, improving the reliability of the results.

Furthermore, a multi-factor scoring mechanism is implemented to combine different evaluation parameters. Instead of relying solely on correctness, the system considers accuracy, confidence scores from the detection model, and behavior analysis results. This ensures a more balanced and meaningful evaluation.

Finally, the system includes a prediction module that analyzes user performance over time. By tracking historical data and identifying patterns, the system can classify the learner's level and provide personalized feedback. This supports continuous improvement and enhances the overall learning experience.

The following sections of this chapter explain each component of the methodology in detail, including system design, model selection, data processing techniques, and evaluation strategies.

2.1 Project Requirements

The Project Requirements define the expected functionalities and system characteristics of the proposed Learning Evaluation and Behavior-Aware Analysis System. These requirements ensure that the system meets user needs while maintaining performance, reliability, and usability standards. The requirements are categorized into functional requirements and non-functional requirements.

2.1.1 Functional Requirements

Functional requirements describe the core operations and services provided by the system.

User Management

- The system shall allow users to register and log in securely.
- The system shall manage user profiles and store user-related data.
- The system shall maintain user performance history for analysis and prediction.

Learning and Evaluation Module

- The system shall allow users to select learning levels (Basic, Intermediate, Advanced).
- The system shall provide quiz-based questions for Basic and Intermediate levels.
- The system shall support multiple Tamil Sign Language categories such as objects, actions, numbers, and expressions.
- The system shall evaluate user responses by comparing them with predefined correct answers.

Video-Based Response Handling

- The system shall allow users to upload video responses for Advanced-level questions.
- The system shall process uploaded videos and extract frames for analysis.

Sign Detection Integration

- The system shall integrate with the sign detection component to identify user-performed signs.
- The system shall receive predicted sign labels and confidence scores from the detection model.
- The system shall validate user responses by comparing predicted outputs with expected answers.

Behavior Analysis

- The system shall analyze user behavior during video-based responses.
- The system shall classify frames into categories such as left, right, up, down, eye open, eye close, and no face.
- The system shall generate an attention score based on aggregated frame-level predictions.

Evaluation and Scoring

- The system shall calculate user scores using a multi-factor scoring mechanism.
- The scoring process shall consider accuracy, confidence score, and behavior (attention score).
- The system shall generate a final score for each learning attempt.

Category-Based Performance Analysis

- The system shall analyze user performance across different sign language categories.
- The system shall calculate category-wise performance scores.
- The system shall identify strong areas and weak areas based on category performance.

Prediction and Feedback

- The system shall classify user learning levels using a rule-based prediction mechanism.
- The system shall analyze performance trends using historical data.
- The system shall generate personalized feedback and improvement recommendations.
- The system shall suggest specific categories for further practice based on weak areas.

Dashboard and Visualization

- The system shall display results through a user dashboard.
- The system shall present final scores, predicted levels, and attention levels.
- The system shall display strong areas, weak areas, and category-wise performance summaries.
- The system shall provide clear and user-friendly visualization of learning progress.

Mentor Support

- The system shall allow mentors or guides to review user performance and progress.
- The system shall enable mentors to provide additional guidance based on identified weak areas.
- The system shall support a hybrid learning approach combining automated feedback and human guidance.

2.1.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints of the system.

Performance

- The system shall process user inputs and generate evaluation results within an acceptable time.
- The system shall support efficient processing of video data for near real-time analysis.

Accuracy

- The system shall maintain high accuracy in behavior detection and evaluation processes.
- The scoring and prediction mechanisms shall produce consistent and reliable results.

Usability

- The system shall provide a user-friendly interface for easy navigation and interaction.
- The system shall ensure that users can complete learning and evaluation tasks without complexity.

Scalability

- The system shall support multiple users simultaneously.
- The system shall handle increasing data volumes efficiently without performance degradation.

Reliability

- The system shall operate consistently without failure during user interactions.
- The system shall ensure correct storage and retrieval of user data.

Security

- The system shall protect user authentication and personal data.
- The system shall ensure secure communication between system components.

Maintainability

- The system shall be designed in a modular architecture to support future enhancements.
- Machine learning models shall be updateable without affecting overall system functionality.

2.2 System Overview

The proposed Learning Evaluation and Behavior-Aware Analysis System is designed as a component of a Tamil Sign Language learning platform to evaluate user performance in a

structured and intelligent manner. The system focuses on assessing user knowledge through multiple evaluation methods, analyzing behavior during learning activities, and providing predictive feedback based on performance trends.

The overall system operates based on an **input–process–output model**. The primary inputs to the system are user interactions, which include quiz responses for Basic and Intermediate levels and video-based responses for the Advanced level. These inputs are processed through several modules, including evaluation, behavior analysis, and prediction, to generate meaningful outputs such as scores, performance insights, and feedback.

At the Basic and Intermediate levels, the system evaluates user knowledge using quiz-based interactions. Users are presented with structured questions related to Tamil Sign Language, and their responses are recorded and analyzed. The evaluation process at these levels is primarily based on correctness, allowing the system to measure the user’s understanding of fundamental concepts and recognition ability.

At the Advanced level, the system introduces a more complex evaluation process where users are required to submit video-based answers. These videos are analyzed using outputs from the sign detection component, which provides the predicted sign and a confidence score. This allows the system to validate whether the user has correctly performed the required sign. Unlike simple quiz-based evaluation, this approach enables real-world assessment of user skills.

In addition to evaluating correctness, the system incorporates a **behavior-aware analysis module**. During video-based responses, frames are extracted from the uploaded video and analyzed to detect the user’s head direction. This is achieved using a deep learning model trained to classify head positions such as left, right, up, down, and straight. The purpose of this module is to determine whether the user is attentive during the response. The results are aggregated across multiple frames to generate an overall attention score, which contributes to the final evaluation.

The system also includes a **multi-factor scoring mechanism** to ensure a more reliable evaluation. Instead of relying solely on correctness, the system combines multiple parameters, including accuracy of the response, confidence score obtained from the sign detection model,

and behavior analysis results. This approach provides a more comprehensive assessment of user performance by considering both the outcome and the learning process.

Furthermore, a **performance tracking and prediction module** is integrated into the system. User performance data, including scores, category-wise results, and historical records, are stored and analyzed over time. Based on this data, the system identifies learning patterns, including strong areas and weak areas across different Tamil Sign Language categories. This allows the system to provide more meaningful and personalized insights rather than only a final score.

The system also generates predictive feedback by analyzing multiple factors such as overall performance, behavior patterns, and category-wise results. In addition to automated feedback, the system supports optional mentor guidance, where instructors or guides can review user progress and provide additional assistance. This hybrid approach enhances the learning experience by combining intelligent system analysis with human support.

Overall, the system is designed to function as an integrated evaluation framework that combines recognition outputs, behavior analysis, learning analytics, and personalized feedback. By doing so, it moves beyond traditional recognition-based systems and provides a more comprehensive and intelligent approach to Tamil Sign Language learning.

2.3 System Architecture

The system architecture of the proposed Learning Evaluation and Behavior-Aware Analysis System is designed to support efficient data processing, modular functionality, and real-time evaluation. The architecture follows a layered approach consisting of the user interface, backend processing modules, machine learning components, and data storage. This design ensures scalability, maintainability, and seamless integration with other components of the Tamil Sign Language learning platform.

The system primarily consists of four main layers: Frontend Layer, Backend Layer, Machine Learning Layer, and Data Storage Layer. Each layer is responsible for specific functionalities, and they work together to provide a complete evaluation process.

2.3.1. Frontend Layer

The frontend layer represents the user interface through which users interact with the system. It is responsible for displaying quizzes, collecting user responses, and allowing users to upload

video-based answers in the Advanced level. The interface is designed to be simple and user-friendly, ensuring that learners can easily navigate through different levels of the system.

At this layer, user inputs are collected in two main forms:

- Quiz responses (Basic and Intermediate levels)
- Video uploads (Advanced level)

These inputs are then sent to the backend for further processing.

2.3.2.Backend Layer

The backend layer acts as the core processing unit of the system. It handles all the application logic, including evaluation, data processing, and communication between different modules. The backend is responsible for receiving user inputs from the frontend and directing them to appropriate processing components.

For quiz-based responses, the backend evaluates answers based on predefined correct values and calculates accuracy. For video-based responses, the backend communicates with the sign detection component to obtain predicted labels and confidence scores. These results are then used in the evaluation process.

The backend also manages the behavior analysis process by extracting frames from uploaded videos and sending them to the machine learning model for classification. Additionally, it handles performance tracking, score calculation, and prediction generation.

2.3.3.Machine Learning Layer

The machine learning layer is responsible for handling intelligent processing tasks within the system. It includes two main components:

1. Sign Detection Component (External Integration)
This component is responsible for detecting the sign performed by the user in the uploaded video. It provides outputs such as predicted labels and confidence scores. These outputs are used by the evaluation system to validate user responses.
2. Behavior Analysis Model (MobileNetV2)
This component is developed as part of the proposed system. It uses a lightweight deep learning model based on MobileNetV2 to classify head direction from video frames. The model predicts whether the user is looking straight, left, right, up, or down. These predictions are aggregated to determine the user's attention level during the response.

The machine learning layer enhances the system by enabling behavior-aware evaluation and improving the reliability of results.

2.3.4.Data Storage Layer

The data storage layer is responsible for storing all relevant data generated by the system. This includes user information, quiz results, video metadata, performance scores, and historical records. The stored data is used for tracking user progress and supporting the prediction module.

By maintaining historical performance data, the system can analyze learning patterns and provide personalized feedback. This layer plays a critical role in enabling learning analytics and prediction capabilities.

2.3.5.System Integration and Data Flow

The overall architecture ensures smooth data flow between different layers. User inputs collected at the frontend are processed by the backend, which interacts with the machine learning layer for advanced analysis. The results are then stored in the database and returned to the frontend as feedback.

This integration allows the system to perform real-time evaluation while maintaining efficiency and accuracy. The modular design also ensures that each component can be improved or updated independently without affecting the entire system.

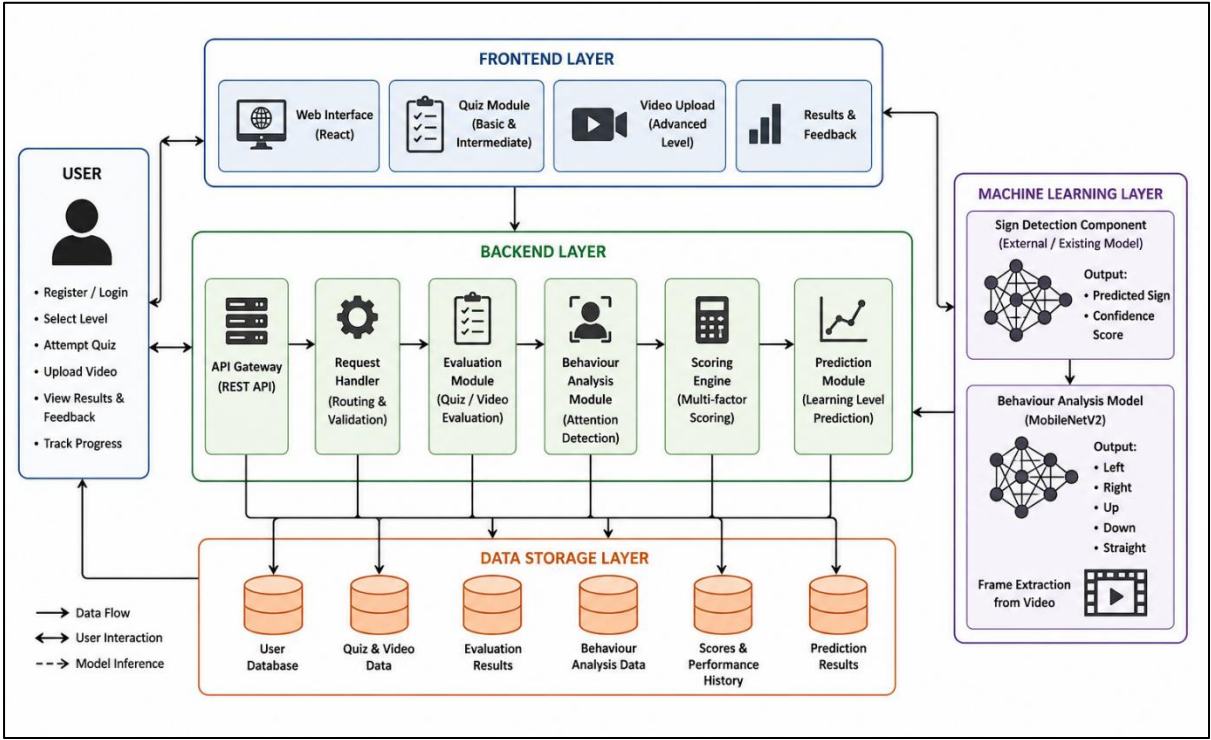


Figure 1: System Architecture of the Learning Evaluation and Behavior-Aware Analysis System

2.4 Methodology

2.4.1 Component Workflow

The component workflow explains how the Learning Evaluation and Behavior-Aware Analysis System processes user activities from the beginning of a learning attempt until the final results and learning insights are generated. The workflow is designed to support three learning levels: Basic, Intermediate, and Advanced, where each level follows a different evaluation method based on the complexity of the learning task.

At the beginning, the user logs into the platform and selects a learning level. In the Basic and Intermediate levels, the user answers quiz-based questions covering multiple Tamil Sign Language categories such as objects, actions, numbers, and expressions. These responses are sent to the backend, where the system compares the selected answers with predefined correct answers. Based on this comparison, the system calculates the user's score and records category-wise performance data.

The Advanced level follows a more detailed workflow, where the user is required to submit a video-based response. After the video is uploaded, the system sends it to the sign detection component. This component identifies the performed sign and returns the predicted label along with a confidence score. The evaluation module then compares the predicted sign with the expected answer to determine correctness.

Simultaneously, the uploaded video is processed by the behavior analysis module. The video is divided into multiple frames, and each frame is passed through the MobileNetV2-based behavior detection model. The model classifies each frame into categories such as left, right, up, down, eye open, eye close, and no face. These frame-level predictions are aggregated to identify user attention patterns and calculate an overall attention score.

After completing quiz evaluation, sign validation, and behavior analysis, the system calculates the final score using a multi-factor scoring mechanism that combines accuracy, confidence score, and attention score. In addition to the overall score, the system organizes user performance data into different sign language categories and calculates category-wise performance scores.

The processed results are then passed to the prediction module. The prediction module analyzes the final score, category-wise performance, behavior patterns, and historical performance data to identify the user's learning level, strong areas, weak areas, and overall engagement level.

Finally, the system displays the results through the dashboard. The dashboard presents detailed outputs, including the final score, predicted learning level, category-wise performance summary, strong and weak areas, attention level, and personalized recommendations for improvement. This workflow ensures that the system evaluates not only correctness but also user behavior, learning patterns, and performance trends, providing a comprehensive and intelligent learning evaluation process.

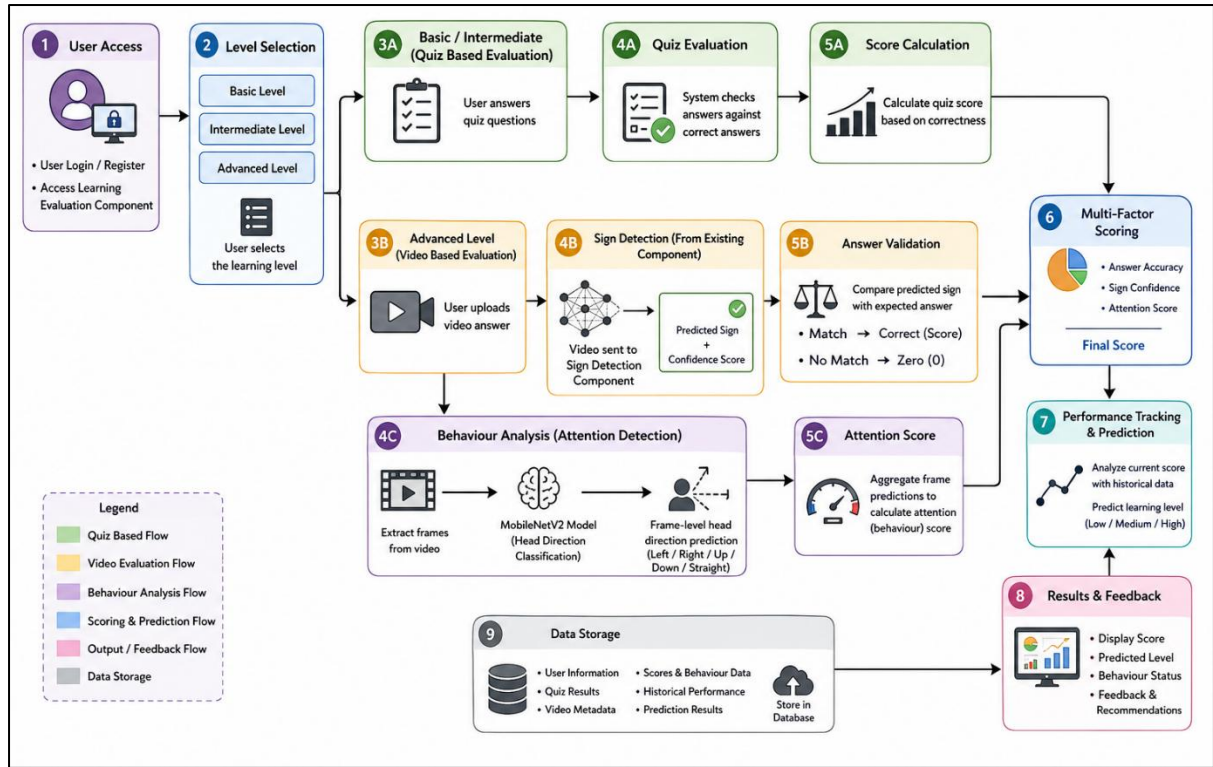


Figure 2: Component Workflow of the Learning Evaluation System

2.4.2 Behavior Detection Model

The Behavior Detection Model is a key component of the proposed Learning Evaluation System, designed to analyze user attention during video-based responses in the Advanced level. Unlike traditional evaluation systems that focus only on correctness, this module introduces an additional layer of analysis by considering user behavior, specifically head direction, to improve the reliability of evaluation results.

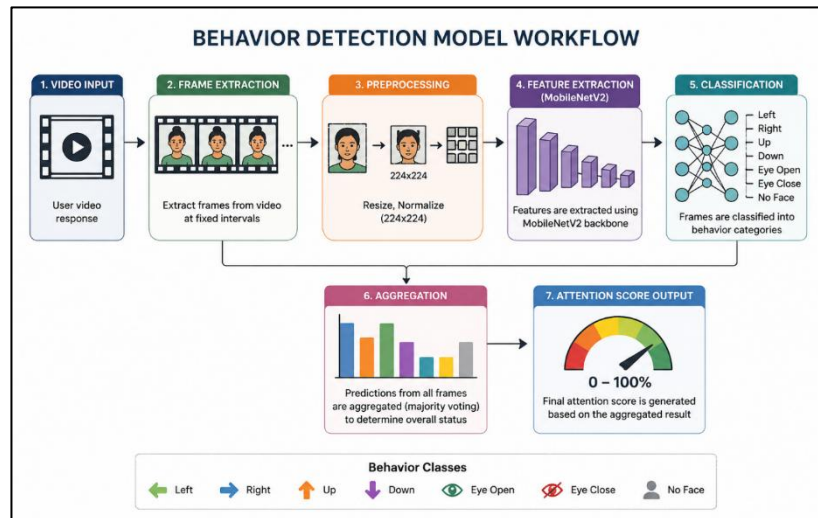


Figure 3: Behavior Detection Model Workflow

2.4.2.1 Model Selection

To perform behavior analysis efficiently, a deep learning-based image classification model was required. In this system, the MobileNetV2 architecture was selected due to its lightweight structure and suitability for real-time applications.

MobileNetV2 is designed using depth wise separable convolutions and inverted residual blocks, which significantly reduce computational complexity while maintaining good classification performance. This makes it an ideal choice for web-based systems where fast processing and low resource usage are important.

The model was implemented using the TensorFlow framework. Unlike transfer learning approaches, the MobileNetV2 model in this system was trained from scratch without using pretrained weights. This allows the model to learn features that are specifically relevant to head direction classification rather than relying on general-purpose image features.

Compared to traditional rule-based approaches such as OpenCV-based face direction estimation, this deep learning model provides better generalization across different lighting conditions, camera angles, and user variations.

2.4.2.2 Dataset Preparation

The behavior detection model was trained using a custom dataset consisting of images representing different head directions. The dataset was organized into seven classes:

- Left
- Right
- Up
- Down
- Eye Open
- Eye Close
- No Face

These classes represent the possible directions a user may look during a video-based response. The dataset was created using a combination of manually collected images and publicly available image sources to ensure diversity in facial orientation, lighting conditions, and backgrounds.

Before training, the dataset was preprocessed by resizing all images to a fixed input size suitable for MobileNetV2 and normalizing pixel values. The dataset was then divided into training and validation sets to evaluate model performance.

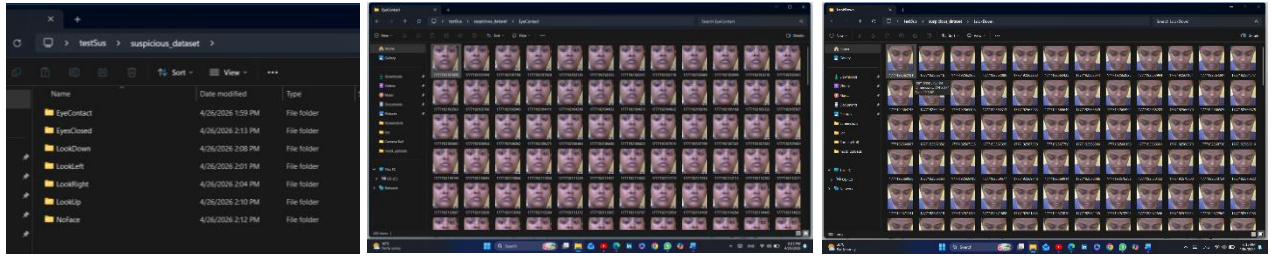


Figure 4: Sample Images from Behavior Detection Dataset

2.4.2.3 Training Process

The model was implemented using **TensorFlow**. During training, the dataset was fed into the MobileNetV2 architecture, which was fine-tuned for classification of head direction.

The following steps were performed:

- Image preprocessing and normalization
- Splitting dataset into training and validation sets
- Training the model using multiple epochs
- Monitoring performance using accuracy and loss

The model was trained using a categorical classification approach, where each image belongs to one of the five predefined classes. A suitable optimizer and loss function were used to ensure stable convergence during training.

2.4.2.4 Model Performance

After training, the model was evaluated using the validation dataset. The performance of the model was analyzed using accuracy and loss metrics, as well as a confusion matrix to evaluate classification performance across different classes.

The training process shows that the model achieved a high level of performance. The training accuracy gradually increased and reached approximately **99%** by the final epoch. The validation accuracy also remained consistently high, reaching approximately **94%–95%**, indicating good generalization of the model.

Although a slight drop in accuracy can be observed around the middle epoch, the model quickly recovered and stabilized, demonstrating robust learning behavior. The loss curves further confirm this, where both training loss and validation loss decrease significantly during initial epochs and stabilize at lower values toward the end of training.

The confusion matrix results show that the model performs well across all classes, with most predictions concentrated along the diagonal. This indicates that the model correctly classifies the majority of input samples. Minor misclassifications are observed between similar directional classes such as "LookLeft" and "LookRight," and between "EyeContact" and "EyesClosed," which is expected due to visual similarity in certain conditions.

Overall, the model demonstrates strong classification performance and is suitable for real-time behavior analysis within the proposed system.

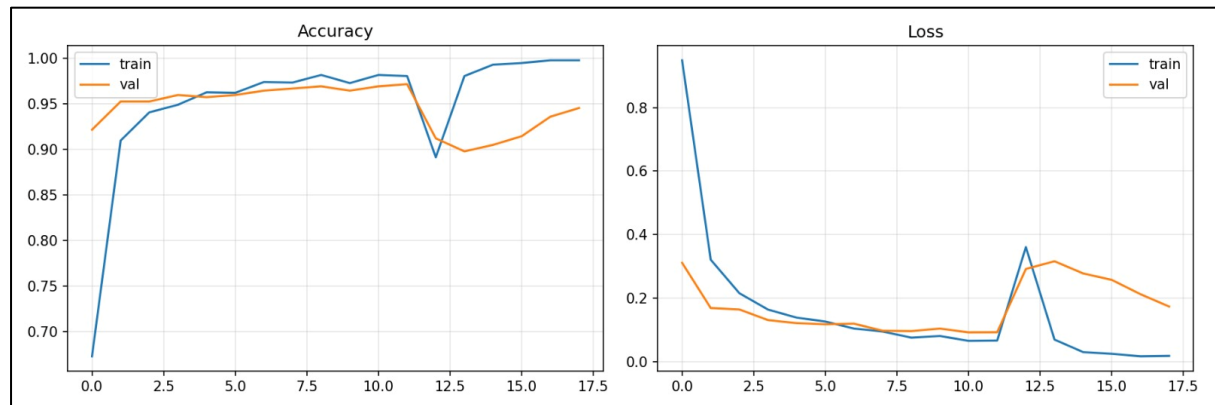


Figure 5: Model Accuracy and Loss Graph

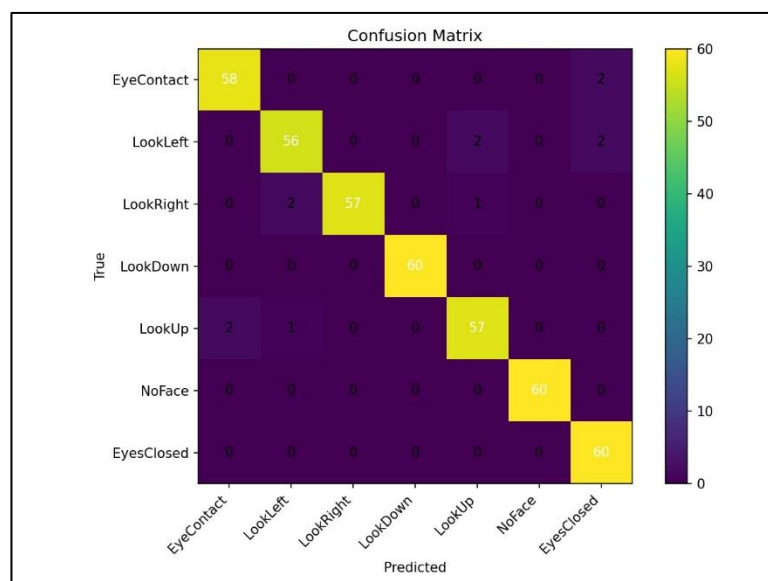


Figure 6: Confusion Matrix of Behavior Detection Model

2.4.2.5 Model Output and Behavior Analysis

During system operation, when a user uploads a video, the video is processed by extracting multiple frames. Each frame is then passed through the trained MobileNetV2 model.

For each frame, the model predicts one of the following classes:

- Left

- Right
- Up
- Down
- Eye Open
- Eye Close
- No Face

These predictions provide detailed insight into user behavior. Direction-based classes (Left, Right, Up, Down) indicate where the user is looking, while Eye Open and Eye Close represent the user's level of engagement. The "No Face" class is used to detect situations where the user is not visible in the frame.

The frame-level predictions are aggregated to determine the overall behavior of the user during the video response. Frames classified as "Eye Open" and forward-facing directions indicate active attention, while frames classified as "Left", "Right", or "Down" may indicate distraction. Frames classified as "Eye Close" or "No Face" are considered strong indicators of disengagement.

Based on these aggregated results, an **attention score** is calculated to represent the level of user engagement during the response.

2.4.3 Evaluation & Scoring Mechanism

The Evaluation and Scoring Mechanism is a core part of the proposed system, responsible for assessing user performance in a structured and meaningful way. Unlike traditional systems that rely only on correctness, this system uses a multi-factor evaluation approach that considers accuracy, model confidence, and user behavior to generate a final score.

2.4.3.1. Evaluation Strategy

The evaluation process is designed based on the three learning levels in the system:

1. Basic and Intermediate Levels

At these levels, user performance is evaluated using quiz-based questions. The system compares user answers with predefined correct answers and calculates the score based on correctness.

This provides a simple and effective way to measure the user's understanding of fundamental concepts.

2. Advanced Level

In the Advanced level, the evaluation process becomes more complex. Users are required to submit video-based answers, which are evaluated using multiple components:

- Sign detection output (predicted sign)
- Confidence score from the detection model

- Behavior analysis (attention score)

This ensures that the evaluation reflects real-world performance rather than simple recognition.

2.4.3.2. Multi-Factor Scoring Mechanism

To improve the reliability of evaluation, the system combines multiple parameters instead of relying on a single factor. The final score is calculated using:

- **Answer Accuracy (A)** → Whether the predicted sign matches the expected answer
- **Confidence Score (C)** → Reliability of the sign detection model
- **Attention Score (B)** → User engagement based on behavior detection

1. Final Score Calculation

The final score is calculated using a weighted multi-factor approach that combines accuracy, confidence, and behavior analysis:

$$Final\ Score = (w_1 \times A) + (w_2 \times C) + (w_3 \times B)$$

Where:

- A = Accuracy (1 for correct answer, 0 for incorrect)
- C = Confidence score obtained from the sign detection model (range: 0–1)
- B = Attention score calculated from the behaviour detection model (range: 0–1)
- w_1, w_2, w_3 = Weight values assigned to each factor

2. Weight Distribution

In this system, the following weight values are used:

- Accuracy (w_1) = 0.5
- Confidence (w_2) = 0.3
- Behavior (w_3) = 0.2

This distribution ensures that **correctness has the highest priority**, while still considering the reliability of the prediction and the user's behavior during the response.

Accuracy is given the highest weight because it directly reflects whether the user performed the correct sign. Confidence score is considered to measure how reliable the prediction is, while behavior analysis contributes to evaluating user engagement.

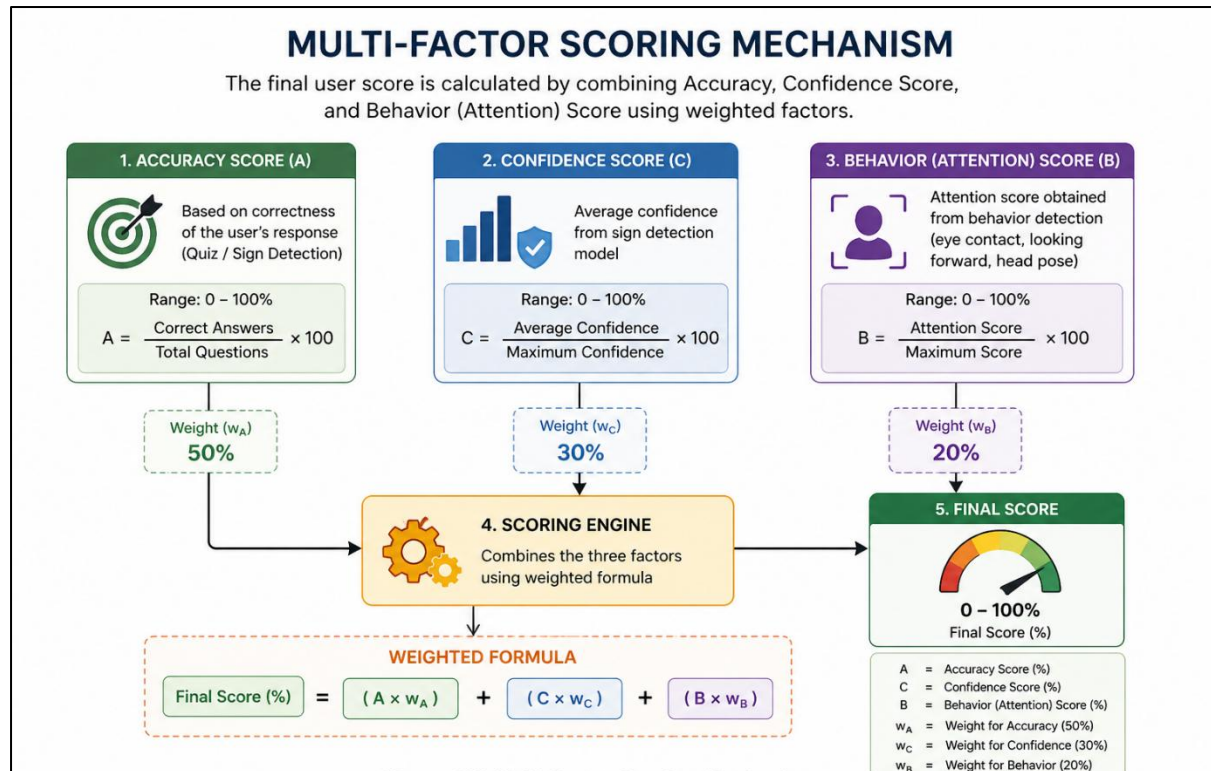


Figure 7: Multi-Factor Scoring Mechanism

2.4.3.3. Behavior-Based Scoring

The attention score is calculated based on predictions generated by the behavior detection model. Each frame of the video contributes to the overall behavior evaluation.

- Frames classified as **Eye Open** → positive attention
- Frames classified as **Left, Right, Up, Down** → partial attention
- Frames classified as **Eye Close or No Face** → low attention

These frame-level results are aggregated to calculate a final attention score between 0 and 1.

2.4.3.4. Performance Tracking

The system stores user scores and evaluation results in the database. This allows tracking of user performance across multiple attempts. By analyzing historical data, the system can identify patterns such as improvement, consistency, or decline.

2.4.3.5. Result Generation

After evaluation, the system generates the following outputs:

- Final Score
- Attention level
- Confidence level

- Strong areas
- Weak areas
- Category-wise performance summary
- Personalized feedback and recommendations

These results are displayed to the user through the dashboard, providing a clear understanding of their performance.

2.4.4 Prediction and Feedback Module

The Prediction and Feedback Module is a core component of the proposed system, designed to analyze user performance in a detailed and structured manner and generate meaningful learning insights. Unlike traditional systems that only provide a final score or correctness feedback, this module evaluates user performance across multiple dimensions, including accuracy, behavior, category-wise performance, and historical trends.

The prediction process follows a **hybrid analytical approach**, where machine learning models are used for feature extraction, and a rule-based mechanism is used for performance interpretation and decision-making. The system does not rely on a separate machine learning model for prediction. Instead, it uses computed evaluation outputs such as accuracy, confidence score, behavior score, and category-wise performance metrics.

Initially, the system collects evaluation data from the scoring module, including final score, attention score, and confidence values. In addition to overall performance, the system organizes user responses into multiple **sign language categories**, such as objects, actions, numbers, and expressions. For each category, the system calculates an average performance score based on user responses.

Using this category-wise analysis, the system identifies:

- **Strong Areas** → categories where the user consistently performs well
- **Weak Areas** → categories where the user shows lower performance or repeated errors

This enables the system to move beyond general scoring and provide targeted insights into the learner's strengths and weaknesses.

In parallel, the system incorporates **behavior-based analysis** using the attention score generated from the behavior detection model. The model evaluates user engagement based on factors such as head direction, eye status, and face presence. This allows the system to understand whether the user was attentive, partially engaged, or distracted during the learning activity.

After collecting all these features, the system applies a **rule-based decision mechanism** to generate predictions. The overall performance score is compared against predefined threshold

values to classify the user's learning level (e.g., beginner, intermediate, advanced). At the same time, category-wise scores and behavior indicators are analyzed to refine the interpretation.

Furthermore, the system utilizes **historical performance data** stored in the database to track user progress over time. By analyzing past attempts, the system can identify learning patterns such as improvement, consistency, or decline. This allows the prediction module to provide more accurate and context-aware insights.

Based on all these factors, the system generates **personalized feedback**, which includes:

- Identification of strong areas
- Identification of weak areas
- Category-wise performance summary
- Attention and engagement level
- Recommendations for improvement
- Suggested categories for further practice

This module transforms raw evaluation data into actionable learning insights, helping users understand not only their score but also how they performed and how they can improve. By integrating scoring, behavior analysis, and category-based prediction, the system provides a comprehensive and intelligent approach to Tamil Sign Language learning.

2.5 Commercialization Aspects of the Product

The proposed Learning Evaluation and Behavior-Aware Analysis System is designed not only as an academic research solution but also as a practical platform that can be extended into a real-world product. With the increasing demand for accessible and technology-driven learning solutions, especially in the field of sign language education, this system has strong potential for commercialization. By combining intelligent evaluation, behavior analysis, and personalized feedback, the system provides value beyond traditional learning tools.

2.5.1 Target Audience

The primary target audience for this system includes individuals who are interested in learning Tamil Sign Language, particularly beginners and intermediate learners who require structured guidance and evaluation. In addition, the system is highly suitable for:

- Students in special education environments
- Hearing-impaired individuals seeking to improve communication skills
- Educational institutions and training centers offering sign language courses
- Teachers and instructors who need tools to monitor and evaluate student progress

Furthermore, the inclusion of performance tracking and feedback mechanisms makes the system useful for self-learning environments, where users can independently assess and improve their skills.

2.5.2 Market Space

The current market for sign language learning platforms is still developing, especially for region-specific languages such as Tamil Sign Language. Most existing solutions focus primarily on sign recognition or static learning content, with limited support for performance evaluation and user engagement analysis.

The proposed system addresses this gap by introducing a more comprehensive approach that combines recognition, evaluation, behavior analysis, and prediction. This creates a unique position in the market as an intelligent learning and assessment platform rather than just a recognition tool.

With the growing adoption of AI-driven education technologies and the increasing focus on inclusive learning, this system has strong potential to expand within educational institutions, online learning platforms, and accessibility-focused applications.

2.5.3 Revenue Earning

The system can be commercialized using several potential revenue models. One of the most practical approaches is a subscription-based model, where users can access advanced features such as video-based evaluation, personalized feedback, and performance tracking through a paid plan.

In addition, the platform can be offered to educational institutions and training centers through licensing agreements, allowing multiple users to access the system under institutional subscriptions. Premium features such as detailed analytics, mentor support, and advanced evaluation modules can also be introduced to generate additional revenue.

Another possible approach is to provide a basic version of the platform for free while offering enhanced features as part of a premium package. This model can help attract a larger user base while maintaining sustainable revenue generation.

Overall, the system has strong potential to evolve into a commercially viable product by leveraging modern digital learning trends and addressing the specific needs of sign language education.

2.6 Testing & Implementation

The testing and implementation phase of the proposed Learning Evaluation and Behavior-Aware Analysis System was conducted to ensure that the system operates correctly, efficiently, and as an integrated solution. This phase includes both the practical implementation of system components and the validation of their functionality through different testing approaches.

2.6.1 Implementation of the Proposed System

The system was implemented as a multi-layered architecture consisting of a frontend interface, backend services, machine learning models, and a database for data storage. The frontend, developed using React.js with Vite, provides an interactive and user-friendly interface where users can navigate through different learning levels, watch lesson videos, answer quiz-based questions, upload video responses, and view their evaluation results. The inclusion of lesson videos allows users to learn Tamil Sign Language concepts before attempting evaluations, thereby improving their understanding and performance.

The backend was developed using a Flask-based API, which acts as the core processing unit of the system. It handles user requests, processes quiz data, manages video uploads, and coordinates communication between different modules such as quiz evaluation, behavior analysis, and prediction generation. The quiz evaluation module was implemented using rule-based logic, where user answers are compared with predefined correct answers. The results are then organized into category-wise performance metrics, enabling the system to identify strong and weak areas.

The behavior detection component was developed using a deep learning model based on MobileNetV2, implemented with TensorFlow and Keras. A custom dataset was created by collecting images representing different user behaviors such as eye contact, looking left, right, up, and down, eyes closed, and no face detection. These images were preprocessed and used to train the model using transfer learning techniques. The trained model was saved in .h5 format and integrated into the system for real-time and video-based analysis.

During execution, uploaded videos are processed frame by frame. Each frame undergoes face detection and preprocessing before being passed to the trained model. A smoothing mechanism is applied using a buffer of recent predictions to improve stability and reduce fluctuations in classification results. The final behavior output is used to compute an attention score, which contributes to the overall evaluation.

In addition, the system integrates external sign detection through an API, which returns predicted sign labels and confidence scores. These outputs are combined with quiz evaluation and behavior analysis to perform multi-factor scoring and generate final results. The system also stores user performance data, enabling further analysis, prediction of learning levels, and generation of personalized feedback.

2.6.2 Testing of the System

To ensure the reliability and accuracy of the system, several testing approaches were carried out, including functional testing, model validation, integration testing, and real-time testing.

Functional testing was performed to verify that all system features operate correctly. This includes testing user interactions such as level selection, quiz submission, video upload, and

result display. The backend APIs were tested using sample inputs to ensure correct data processing and response generation.

The behavior detection model was validated during the training phase using a validation dataset. Performance metrics such as accuracy and loss were analyzed to confirm that the model learned effectively and generalized well to unseen data. The model demonstrated consistent performance in classifying different user behaviors.

Integration testing was conducted to ensure that all system components work together seamlessly. This includes the interaction between the frontend interface, backend API, machine learning model, and external sign detection service. The system successfully processed both quiz-based and video-based inputs and generated complete evaluation results without errors.

Real-time testing was also performed using webcam input and recorded video samples. These tests were used to evaluate how the system responds to different user behaviors such as attentive and distracted states. The results confirmed that the system can reliably detect behavior patterns and incorporate them into the evaluation process.

The prediction and feedback module was tested using multiple user scenarios. The system was able to identify strong areas and weak areas based on category-wise performance and generate meaningful recommendations. The integration of accuracy, confidence, and behavior ensured that the evaluation reflects both correctness and user engagement.

Testing Type	Description	Method Used	Result
Functional Testing	Verified quiz evaluation, video upload, and result display	Manual testing with sample inputs	Passed
Model Testing	Evaluated behavior detection model performance	Validation dataset and training metrics	Accurate
Integration Testing	Tested interaction between frontend, backend, and ML modules	End-to-end testing	Successful
Real-Time Testing	Tested behavior detection using live webcam and video inputs	Live testing scenarios	Reliable
Prediction Testing	Verified strong/weak area detection and feedback generation	Scenario-based testing	Consistent

Table 2 : Summary of Testing Activities

2.6.3 Summary of Implementation and Testing

Overall, the implementation and testing phase confirms that the system functions as a reliable and integrated platform. The combination of rule-based evaluation, machine learning-based behavior analysis, and prediction mechanisms enables the system to generate accurate and meaningful results.

The testing results demonstrate that the system can effectively process user inputs, analyze behavior, and provide personalized feedback in a consistent manner. This validates the effectiveness of the proposed approach and its suitability for real-world learning applications.

2.7 Tools and Technologies

The development of the proposed Learning Evaluation and Behavior-Aware Analysis System involved the use of several tools and technologies across different layers, including frontend development, backend processing, machine learning, and data handling. These technologies were selected based on their efficiency, flexibility, and suitability for building an integrated intelligent system.

1) Frontend Technologies

The frontend of the system was developed using **React.js** with **Vite** as the build tool. React was selected due to its component-based architecture, which allows efficient development of dynamic and reusable user interface elements. It enables the system to manage complex interactions such as quiz rendering, video uploads, and result visualization in a structured manner.

Vite was used to enhance the development experience by providing faster build times and efficient module handling. Its support for hot module replacement allows quick updates during development, improving productivity and performance.

2) Backend Technologies

The backend of the system was implemented using **Flask**, a lightweight Python web framework. Flask was chosen due to its simplicity and flexibility in building RESTful APIs. It allows seamless integration with machine learning models and supports efficient handling of HTTP requests, including quiz data submission and video uploads.

The backend also facilitates communication between different system components, making it suitable for managing the overall workflow of the application.

3) Machine Learning and AI Technologies

The behavior detection component was developed using **TensorFlow** and **Keras**, which are widely used frameworks for deep learning applications. TensorFlow provides a powerful computational backend for model training and execution, while Keras offers a high-level interface that simplifies the process of building and training neural networks.

The system utilizes **MobileNetV2** as the base model for behavior classification. This model was selected due to its lightweight architecture and ability to perform efficiently in real-time applications without requiring high computational resources.

4) Data Handling and Processing

The system uses **NumPy** for numerical computations and array operations, which are essential for handling image data and model inputs. **Pandas** is used for organizing and processing structured data, particularly in the analysis and prediction modules.

These libraries contribute to efficient data manipulation and feature processing within the system.

5) Model Training and Visualization Tools

During the training phase, additional tools such as **Matplotlib** were used to visualize model performance through accuracy and loss graphs. This helps in understanding the learning behavior of the model and evaluating its performance over time.

6) Development Environment

The system was developed using **Python** as the primary programming language, due to its strong support for machine learning and backend development. The development environment included standard tools and libraries that support rapid development, testing, and integration of different system components.

7) Database Technologies

The system uses Firebase as the database for storing and managing application data. Firebase was selected due to its real-time data handling capabilities and easy integration with web applications.

3 RESULTS AND DISCUSSION

The results of the proposed Learning Evaluation and Behavior-Aware Analysis System demonstrate the effectiveness of the system in evaluating user performance and analyzing behavior during the learning process. The system was tested using both quiz-based inputs and video-based responses to validate its functionality under different scenarios.

For the Basic and Intermediate levels, the system successfully evaluated user responses by comparing them with predefined correct answers. The results were calculated accurately and displayed through the user interface. In addition to overall scores, the system was able to

generate category-wise performance results, allowing users to understand their strengths and weaknesses across different Tamil Sign Language categories.

For the Advanced level, the system processed user-uploaded videos and performed both sign detection and behavior analysis. The behavior detection model was able to classify user actions such as eye contact, looking left, right, up, down, eyes closed, and no face detected. These classifications were aggregated to determine the user's attention level during the response.

The system also generated a final evaluation score by combining multiple factors such as answer accuracy, confidence scores from sign detection, and behavior-based attention scores. This multi-factor approach ensured that the evaluation reflects not only correctness but also user engagement.

Furthermore, the prediction module analyzed user performance data and successfully identified strong areas and weak areas based on category-wise results. The system was able to generate meaningful feedback and recommend areas for improvement, which enhances the overall learning experience.

3.1 Behavior Detection Results

The behavior detection component was tested using both recorded video inputs and real-time webcam data. The results show that the system can effectively distinguish between different user behavior patterns.

When the user maintains eye contact and looks directly at the screen, the system correctly identifies the behavior as attentive. In contrast, when the user looks away, such as to the left, right, upward, or downward, the system detects reduced attention levels. Additionally, cases where the user closes their eyes or moves out of the camera frame are also correctly classified.

These behavior patterns are detected using a combination of deep learning predictions and rule-based analysis, including head pose estimation and eye aspect ratio calculations. The results confirm that the system can reliably monitor user engagement during the evaluation process.

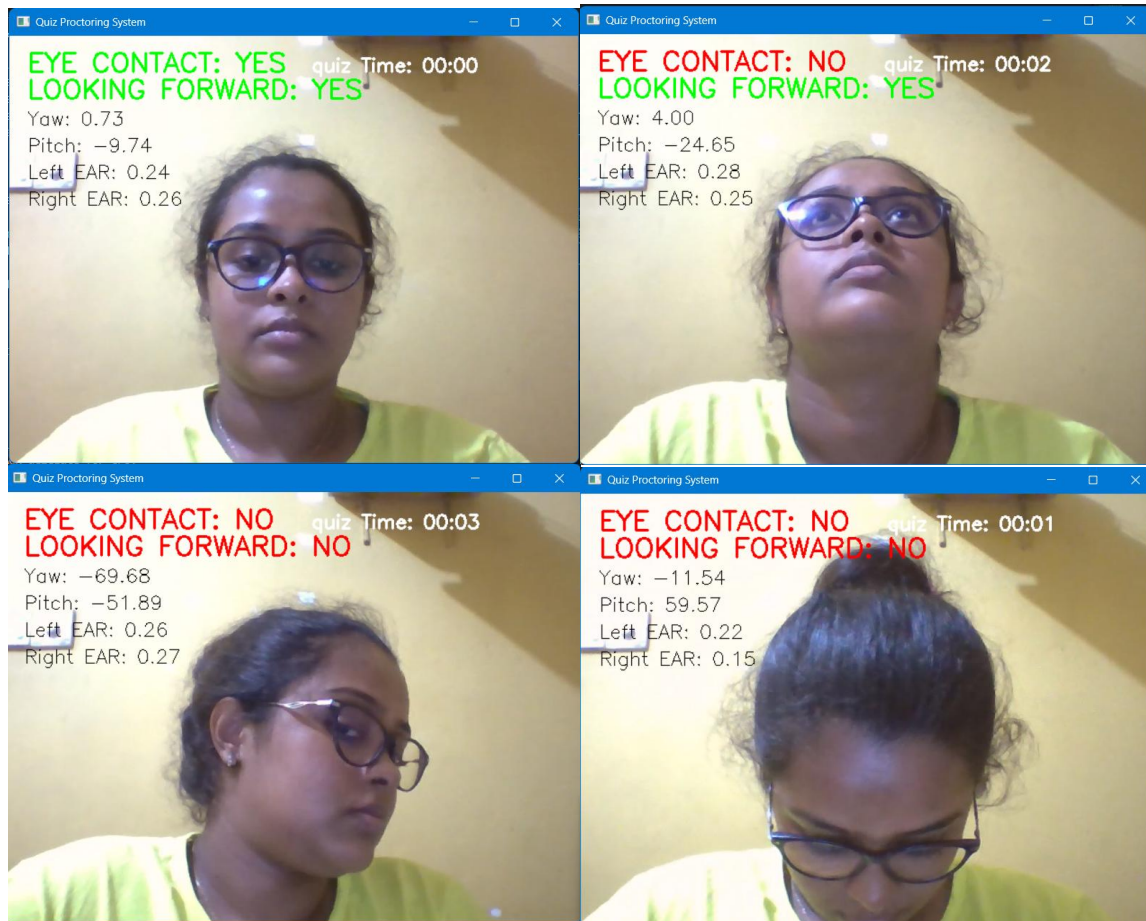


Figure 8 : Behavior Analysis Outputs

3.2 Prediction and Evaluation Results

The prediction module was tested using multiple user scenarios to evaluate its ability to generate meaningful insights. Based on the user's performance data, the system successfully identified strong areas and weak areas across different categories.

The system also classified users into different learning levels based on their performance scores and behavior analysis. The feedback provided by the system included recommendations for improvement, helping users focus on areas where they need more practice.

The results indicate that the integration of category-based analysis and behavior detection improves the quality of evaluation compared to traditional systems that rely only on correctness.

3.3 Discussion

The results obtained from the proposed Learning Evaluation and Behavior-Aware Analysis System highlight the effectiveness of combining multiple evaluation factors to improve the overall assessment process. Unlike traditional learning systems that rely only on correctness, this system integrates accuracy, behavior analysis, and prediction mechanisms to provide a more comprehensive evaluation.

One of the key strengths observed in this system is the integration of behavior detection with the evaluation process. By analyzing user behavior through video inputs, the system is able to determine the level of attention during the learning activity. This adds an additional dimension to the evaluation, as it captures not only what the user answers, but also how the user behaves while answering. The experimental results show that this approach helps in identifying cases where users may provide correct answers but exhibit low engagement, which would not be detected in conventional systems.

The behavior detection model demonstrated satisfactory performance in classifying different user states such as eye contact, head movement, and eye closure. The use of MobileNetV2 with transfer learning allowed the model to achieve stable and reliable predictions even with a relatively small custom dataset. Additionally, the incorporation of smoothing techniques helped reduce fluctuations in frame-level predictions, resulting in more consistent behavior analysis.

Another important aspect of the system is the use of category-wise performance analysis. Instead of providing only an overall score, the system evaluates user performance across different Tamil Sign Language categories. This enables the identification of strong areas and weak areas, which is essential for personalized learning. The results show that users can gain a clearer understanding of their learning progress and focus on specific areas that require improvement.

The prediction and feedback module further enhances the system by generating meaningful insights based on user performance. By analyzing historical data and category-level results, the system is able to classify user learning levels and provide targeted recommendations. This makes the system more interactive and supportive compared to traditional evaluation methods. The inclusion of explainable elements in the prediction process also improves transparency, allowing users to better understand how their results are generated.

The integration of multiple components, including quiz evaluation, sign detection, behavior analysis, and prediction, demonstrates the effectiveness of a hybrid approach. Each component contributes to the final outcome, and their combined effect results in a more reliable and intelligent evaluation system. The successful interaction between these components indicates that the system architecture is well-designed and capable of handling complex workflows.

However, some limitations can be identified in the current implementation. The behavior detection model depends on the quality of video input and lighting conditions, which may affect accuracy in real-world scenarios. Additionally, the dataset used for training is relatively limited, which may restrict the model's ability to generalize to a wider range of user behaviors. Future improvements can include expanding the dataset, incorporating more diverse samples, and enhancing the robustness of the model.

Overall, the discussion confirms that the proposed system provides a more advanced and effective approach to learning evaluation by combining machine learning, rule-based analysis, and user behavior monitoring. The system not only evaluates correctness but also considers user engagement and learning patterns, making it suitable for real-world educational applications.

4 CONCLUSIONS

The proposed Learning Evaluation and Behavior-Aware Analysis System presents an effective and intelligent approach for evaluating Tamil Sign Language learning. The system was designed to address the limitations of traditional evaluation methods by incorporating multiple factors such as answer accuracy, user behavior, and performance-based prediction.

Through the implementation of quiz-based evaluation and video-based analysis, the system successfully evaluates users across different learning levels. The integration of a behavior detection model enables the system to monitor user attention during responses, adding an important dimension to the evaluation process. This ensures that the assessment is not limited to correctness alone but also reflects user engagement.

The use of category-wise performance analysis further enhances the system by identifying strong and weak areas in a structured manner. This allows users to better understand their learning progress and focus on areas that require improvement. In addition, the prediction and feedback module provides personalized recommendations, making the system more supportive and user-centered.

From a technical perspective, the system demonstrates the successful integration of multiple technologies, including machine learning, and web-based application development. The behavior detection model, developed using MobileNetV2, showed satisfactory performance in classifying different user behaviors. The backend system effectively handled data processing, evaluation, and communication between different modules.

Overall, the results confirm that the proposed system functions as a reliable and efficient platform for learning evaluation. By combining automated analysis with structured feedback, the system provides a more comprehensive learning experience compared to conventional approaches.

For future improvements, the system can be enhanced by expanding the dataset used for behavior detection, improving model accuracy under varying real-world conditions, and incorporating additional features such as more advanced prediction techniques and extended learning content. These enhancements would further strengthen the system's applicability in real-world educational environments.

In conclusion, the proposed system demonstrates a practical and scalable solution for intelligent learning evaluation, with the potential to support effective Tamil Sign Language education and improve user learning outcomes.

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